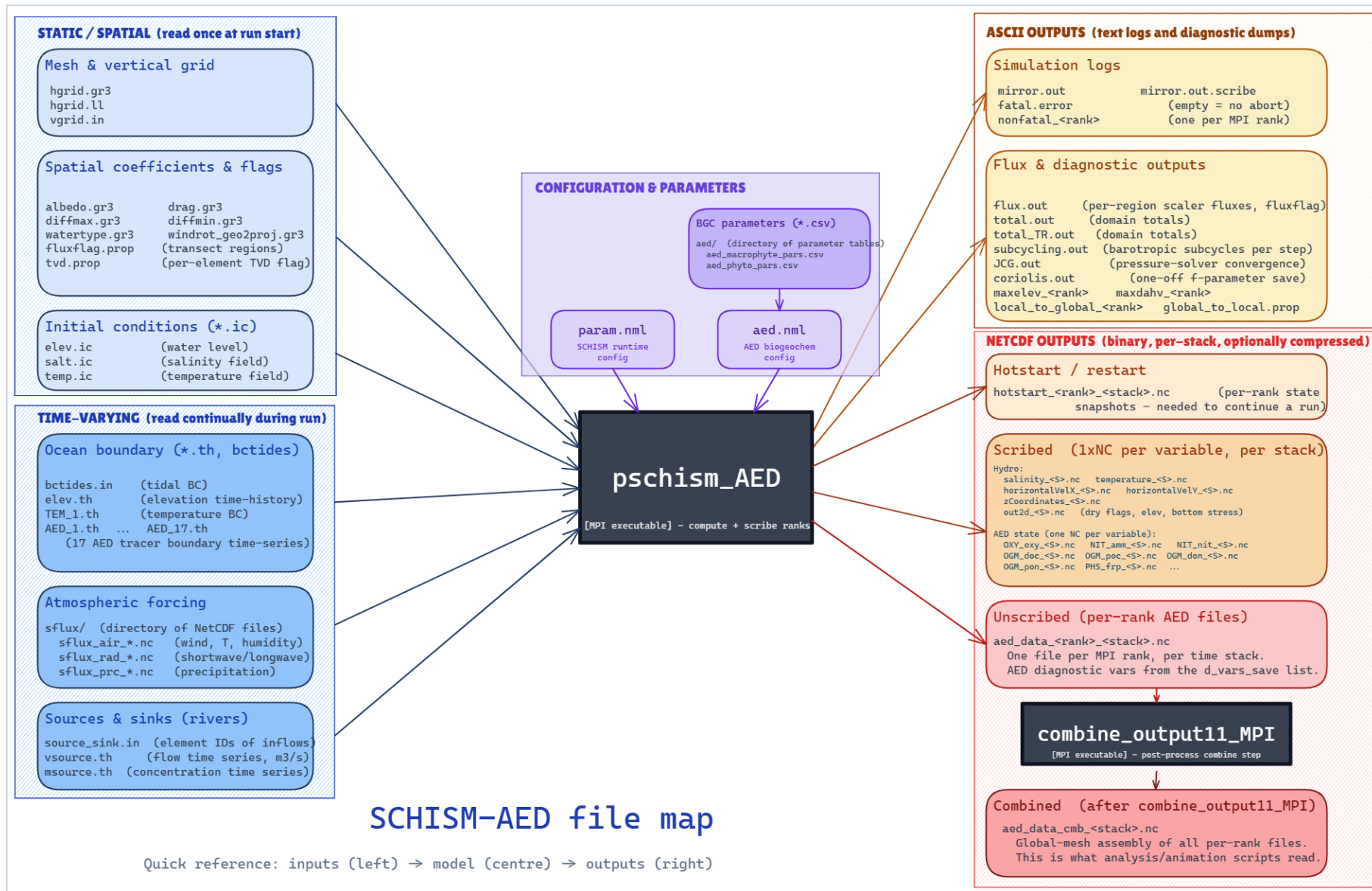


SCHISM-AED file table



Quick reference for the files involved in a SCHISM-AED run. Sub-headings within each table mirror the box names in the diagram above.

Executables

File	Description	Notes
<code>pschism_AED</code>	Main SCHISM-AED MPI executable. Reads <code>param.nml</code> + <code>aed.nml</code> , advances the model, writes hydro + AED outputs.	Launched with <code>mpirun -np N pschism_AED <NSCRIBE></code> . NSCRIBE = ranks reserved for scribe-I/O; compute ranks = N – NSCRIBE.
<code>combine_output11_MPI</code>	Post-process binary that assembles per-rank AED output files into a single global-mesh file.	Runs after <code>pschism_AED</code> . Usage: <code>mpirun -np 4 combine_output11_MPI -b <begin> -e <end> -o aed_data .</code> Reads <code>local_to_global_<rank> maps</code> .

Inputs

File	Description	Notes
CONFIGURATION — purple boxes above the model		
<code>param.nml</code>	SCHISM runtime configuration namelist.	Time step, run length, output flags (<code>iof_hydro</code>), solver options, flux options (<code>iflux</code>), I/O cadence.
<code>aed.nml</code>	AED biogeochemistry configuration namelist.	Active AED modules, state variables, diagnostics to save (<code>d_vars_save</code>), output compression level.
<code>aed/aed_macrophyte_pars.csv</code> <code>aed/aed_phyto_pars.csv</code>	BGC parameters (*.csv) — per-group parameter tables for the macrophyte and phytoplankton modules.	Read by <code>aed.nml</code> when those modules are active. One row per group/strain.

File	Description	Notes
STATIC / SPATIAL — read once at run start		
Mesh & vertical grid		
<code>hgrid.gr3</code>	Horizontal unstructured mesh: node coordinates (projected/metric), depths, element connectivity.	The model's geometric backbone — every other input/output is indexed off this mesh.
<code>hgrid.ll</code>	Same mesh in longitude/latitude coordinates.	Used by sflux interpolation and any spherical-coordinate calcs.
<code>vgrid.in</code>	Vertical coordinate configuration.	Defines sigma layers / LSC2 / hybrid scheme. For a 2D run typically 2 layers.
Spatial coefficients & flags		
<code>albedo.gr3</code>	Surface albedo, per node.	Used in the radiation balance for heat-flux calcs.
<code>drag.gr3</code>	Bottom-drag coefficient, per node.	Spatially-varying bottom friction.
<code>diffmax.gr3</code> / <code>diffmin.gr3</code>	Maximum / minimum horizontal diffusivity, per node.	Cap and floor for <code>Dx</code> / <code>Dy</code> in the diffusion step.
<code>watertype.gr3</code>	Jerlov water type / light-extinction class, per node.	Controls light penetration in the radiation model.
<code>windrot_geo2proj.gr3</code>	Per-node rotation to convert geographic wind to projected-coords wind.	Bridges sflux (geographic) and model (projected) coordinate systems.
<code>fluxflag.prop</code>	Per-element flag identifying flux-transect regions (–1..N).	Required for <code>iflux=2</code> boundary-flux accounting → contents of <code>flux.out</code> .
<code>tv.d.prop</code>	Per-element TVD on/off flag.	Selects which elements use the Total-Variation-Diminishing transport scheme.
Initial conditions (*.ic)		
<code>elev.ic</code>	Initial water-level field.	Per-node initial elevation.
<code>salt.ic</code>	Initial salinity field.	Per-node initial salinity (PSU).
<code>temp.ic</code>	Initial temperature field.	Per-node initial temperature (°C).

File	Description	Notes
TIME-VARYING — read continually during the run		
Ocean boundary (*.th, bctides)		
bctides.in	Tidal boundary configuration: nodes, constituents, amplitudes, phases.	Defines which open boundaries are tidal and how they're driven.
elev.th	Time series of elevation at open boundaries.	Time column = seconds since <code>start_year/month/day</code> .
TEM_1.th	Time series of temperature at boundary segment #1.	One file per active TEM boundary segment.
AED_1.th ... AED_17.th	Time series of AED tracer concentrations at each boundary segment.	One file per AED-tracer boundary segment (here 17 segments). Order matches <code>aed_models</code> in <code>aed.nml</code> .
Atmospheric forcing		
sflux/sflux_air_*.nc	NetCDF atmospheric forcing — wind components, air temperature, specific humidity.	One file per period (typically daily or stack-sized).
sflux/sflux_rad_*.nc	NetCDF atmospheric forcing — shortwave / downward longwave radiation.	Same period structure as <code>sflux_air</code> .
sflux/sflux_prc_*.nc	NetCDF atmospheric forcing — precipitation.	Optional — present only if precipitation is needed.
Sources & sinks (rivers)		
source_sink.in	List of mesh element IDs where rivers / point sources enter.	This run has 14 sources; source #3 is Pioneer River main gauge. See <code>FLOOD.md</code> for the full mapping.
vsources.th	Hourly flow time series (m ³ /s) for each source element.	Column 1 = time (seconds), columns 2...N+1 = flow at each source element.
msources.th	Hourly concentration time series for each source × tracer combination.	Larger than <code>vsources</code> (one column per source × tracer).

Outputs

File	Description	Notes
ASCII OUTPUTS — text logs and diagnostic dumps		
Simulation logs		
<code>mirror.out</code>	Main runtime log — step counts, dt, mass-conservation diagnostics, init/finalize.	Rank 0 only. First place to check after a run.
<code>mirror.out.scribe</code>	Companion log from the dedicated scribe-I/O rank.	Contains the full scribed-variables table (<code>iout_23d</code> per variable).
<code>fatal.error</code>	Final abort message if <code>parallel_abort</code> was called.	0 bytes = no fatal abort. Single rank-0 file.
<code>nonfatal_<rank></code>	Per-rank non-fatal warnings (e.g. Kriging matrix info).	One file per MPI rank. <code>grep</code> -able en masse.
Flux and diagnostic outputs		
<code>flux.out</code>	Per-region tracer transports across boundaries defined in <code>fluxflag.prop</code> .	Active when <code>iflux=1</code> or <code>iflux=2</code> . Drives boundary-flux mass balance (see <code>schism_run_process_3.ipynb</code> cell 42+).
<code>total.out</code>	Domain-integrated hydrodynamic totals per output step (volume, kinetic energy).	Rank 0 only.
<code>total_TR.out</code>	Same as <code>total.out</code> but for tracer state variables.	Cheapest sanity check for AED tracer mass balance.
<code>JCG.out</code>	Pressure-solver convergence log: iterations + residual norms per step.	Diagnoses solver stalls / bad time steps.
<code>coriolis.out</code>	One-off per-node dump of (lon, lat, Coriolis f).	Verification of the f-parameter field; no time dimension.
<code>subcycling.out</code>	Per-step count of barotropic subcycles needed (1 = none).	Stability monitor — max value flags the most stressed step.
<code>maxelev_<rank></code>	Per-rank running maximum elevation at each node.	Inundation envelope — useful for storm-surge analysis.
<code>maxdahv_<rank></code>	Per-rank running maximum depth-averaged horizontal velocity at each node.	Useful for scour / sediment-transport analysis.
<code>local_to_global_<rank></code> <code>global_to_local.prop</code>	Per-rank partition map (local → global node/element IDs) and its inverse.	Required by <code>combine_output11_MPI</code> . Not data per se but indispensable.

File	Description	Notes
NETCDF OUTPUTS — binary, per-stack, optionally compressed		
Hotstart / restart		
hotstart_<rank>_<stack>.nc	Per-rank snapshots of the full model state (hydro + AED).	Needed to continue a run from a checkpoint. Written at intervals set by <code>nhot_write</code> in <code>param.nml</code> . Combine with <code>combine_hotstart7</code> before reuse.
Scribed (one NC per variable, per stack)		
out2d_<S>.nc	2D fields aggregated into a single NC per stack: dry-cell flags, elevation, bottom stress, depth-averaged velocity, plus a couple of element-centred AED state vars.	Always written. Stack <code><S></code> is the time-chunk number.
salinity_<S>.nc	Salinity (3D, node-centred).	<code>iof_hydro(19)=1</code> .
temperature_<S>.nc	Water temperature (3D, node-centred).	Requires <code>iof_hydro(18)=1</code> .
horizontalVelX_<S>.nc horizontalVelY_<S>.nc	X / Y components of horizontal velocity (3D, node-centred).	<code>iof_hydro(16)=1</code> .
zCoordinates_<S>.nc	Z-coordinate of each layer/node (3D).	Needed for any vertical-profile analysis.
OXY_oxy_<S>.nc , NIT_amm_<S>.nc , NIT_nit_<S>.nc , OGM_*_<S>.nc , PHS_frp_<S>.nc , ...	One file per AED state variable, 3D node-centred.	Scribed automatically whenever AED is on. Filename = variable name.
Unscribed (per-rank AED files)		
aed_data_<rank>_<stack>.nc	Per-MPI-rank, per-stack NetCDF with AED diagnostic variables (the <code>d_vars_save</code> list from <code>aed.nml</code>).	Written directly by <code>pschism_AED</code> ; element/face-centred. Must be assembled via <code>combine_output11_MPI</code> before analysis. With deflate enabled (<code>nc_deflate_level=4</code>), each file is ~10–15 MB.
Combined (after <code>combine_output11_MPI</code>)		
aed_data_cmb_<stack>.nc	Global-mesh assembly of all per-rank AED files for one stack.	What every analysis / animation script reads. Contains both 3D pelagic diagnostics (e.g. <code>OXY_sat</code> , <code>NIT_nitrif</code>) and 2D sheet diagnostics (e.g. <code>OXY_oxy_dsf</code> , <code>OGM_pon_swi</code>). Stack-1 file is typically 400–500 MB after compression.

Quick start guide

1. Building from source

On a fresh Ubuntu host, the full SCHISM-AED build flow is:

```
git clone https://github.com/AquaticEcoDynamics/AED_Tools
cd AED_Tools/
./fetch_sources.sh schism          # pull the SCHISM source tree
./fetch_sources.sh all             # pull the AED libraries and helpers
sudo apt install libopenmpi-dev
sudo apt install libnetcdf-dev libnetcdf-dev
ln -s schism a                    # convenience symlink for the patch
patch -p0 < schism-aed/aed-schism.xdiff
cd schism/src
ln -s ../../schism-aed/src/AED .   # graft AED into SCHISM's source tree
cd ../../
./build_schism.sh --with-aed       # compile pschism_AED + combine_output11_MPI
ls binaries/ubuntu/22.04/         # confirm the executables landed here
```

2. Running the model

Place `param.nml`, `aed.nml`, all `*.gr3` / `*.ic` / `*.th` / `bctides.in` inputs and the `sflux/` forcing directory in a single run directory. Then either invoke `mpirun` directly:

```
# In the run directory:
mpirun -np 64 pschism_AED 24
# 64 ranks total; last integer is NSCRIBE
# (ranks reserved for scribe-I/O).
# Compute ranks = NPROC - NSCRIBE = 40.
```

...or use the wrapper script which also runs the post-process combine step in one go:

```
./run_schism_vm.sh
# Override defaults via env vars, e.g.:
NPROC=32 NSCRIBE=8 COMBINE_END=14 ./run_schism_vm.sh
```

3. At-a-glance run health check

After a run completes, the cheap sanity-check sequence is:

```
cd outputs
cat fatal.error
# should be empty (0 bytes = no fatal abort)
tail mirror.out
# should end with a clean finalization message
grep -i error nonfatal_*
# quick scan for rank-local warnings
ls aed_data_cmb_*.nc
# confirm combined AED files exist
```

If those look right, the model ran cleanly and the data is ready to analyse.